Final Portfolio

Light & Sight '23





RANDOLPH-MACON

COLLEGE

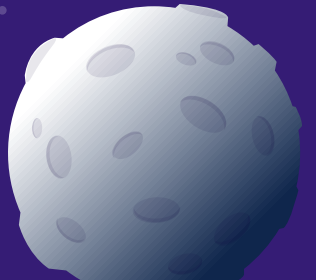






Physics of Light Elements





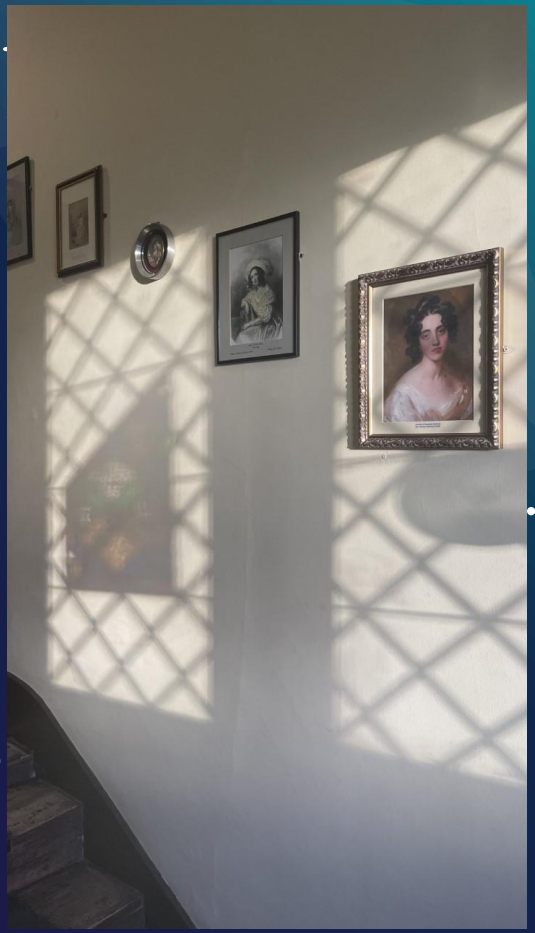


Shadows

Spatial Coherence

Shadows are an effect that occurs at the whim of the light source and the object that is blocking the light from traveling through. Opaque objects block all light, while translucent materials allow some light to pass and transparent materials stops very little, if any at all. When it comes to the light source, the spatial coherence can change the intensity of a shadow. Spatial coherence is the density and direction of the light; this is seen primarily in light beams and lasers, unlike sunlight and lightbulbs.

As one can see, both images are subjected to very little spatial coherence. This is because the light source that is creating the shadow is sunlight, which travels in all directions. Another notable feature of the shadows in these pictures are the “objects”. The “objects” on the left are people, therefore this demonstrates opacity and light is blocked as a result. On the right, stained glass windows are shown, illustrating transparency. Due to the colorful nature of the window, the shadows have a hue instead of an absence of light or white light.





Reflection

Convex Mirror

- When a photon bounces off a surface, an image is contrived in a likeness to the "real" object. The direction of light changes, but not its position or location on the material. This is called reflection, which can occur on surfaces like mirrors or areas with low albedo. Mirrors come in all different shapes and sizes but can generally be compartmentalized into three categories: plane mirrors, convex mirrors, and concave mirrors. All mirrors create a reflection with opposite handedness; what truly determines the difference between these three variations is the level of distortion.

- The image to the right is a regular plane mirror, while the image to the left is a very cylindrical convex mirror. Due to the mirror's curvature, light rays hit the surface at difference points. According to the Law of Reflection, the angle of incidence is equal to the angle of reflection. Therefore, when both elements are taken into consideration (Law of Reflection and curvature of a surface), these concepts can be manipulated into the illusion seen on the left. The "real" image is contorted in such extremity that the reflected image appears to be "normal" (a picture of a king is visible).



Refraction

Snell's Law



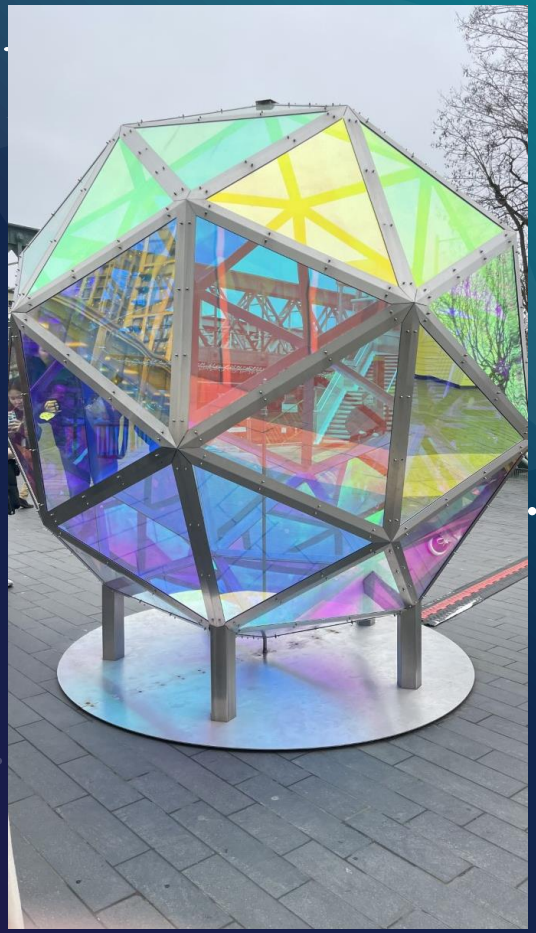
- Unlike its counterpart reflection, refraction is a consequence of light rays bending when transitioning from one transparent medium to another. Snell's Law, or the Law of Refraction, states that the two mediums forces the angle between the exiting ray and the line perpendicular to the surface to be related to the angle of the incoming ray. The Refractive Index is dependent on the resistance of a medium's atoms to movement caused by light and furthermore, measures the speed as to which light passes through said medium.
- The picture is situation where two objects are placed inside a transparent, solid container that is filled with a clear liquid. At first glance, it appears that an object in the liquid is "broken". This is not a correct conclusion; because of Snell's Law, there is only the illusion of a broken straw because of the refraction (shape of container and aligment of mediums).

Polarization

Birefringence

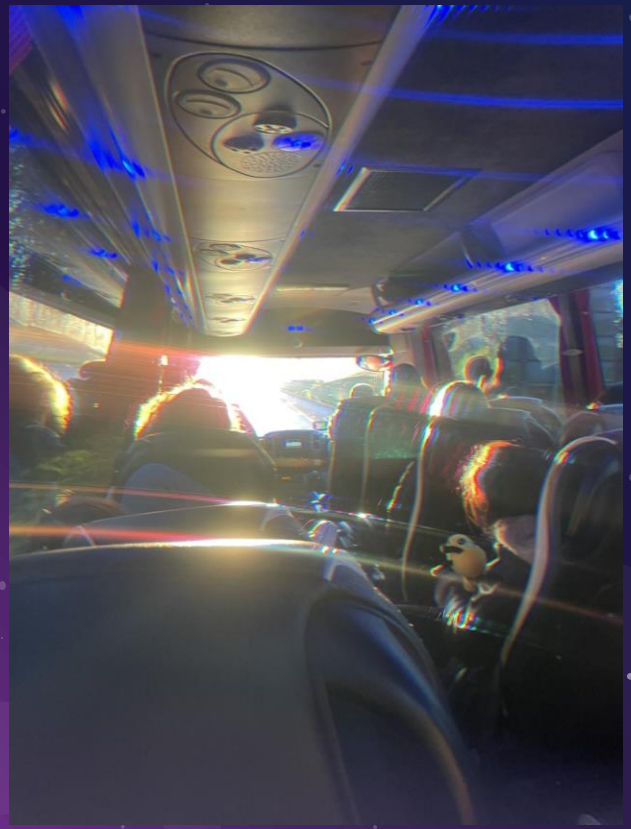


- Light is an electromagnetic wave; as a wave, there are vertical and horizontal oscillations at right angles (transverse). The direction at which the electric field follows is called polarization, and the intersection between the vertical oscillating polarized wave and the horizontal counterpart generates a phenomenon called birefringence. Birefringence can be seen in many different surfaces, such as tape, glass, film, etc. While gazed upon, the birefringent material appears colorful and can change color as the observer moves around the object.
- The picture to the left proves the phenomenon with the colorful exterior of the shaped glass. As the observer walks by the lantern, various new sections of the glass adopts the bright, rainbow-like appearance. This is due to the lack of light absorption and instead, there is a change in the polarization of light. This explains why the color shifts as the observer perceives at new angles: the birefringent material has a special orientation, so the atoms are arranged in such a way that there is an added preferential direction.



Interference & Diffraction

Diffraction



- Diffraction is an effect that happens when light beams slowly diverge as they propagate. The divergence is an outcome from the light waves propagating in more than one direction. When the light diffracts, patterns form and interference fringes are made (diffraction pattern). These patterns are typically colorful and presents itself in shapes of long streaks.
- This picture was captured on a bus, where sunlight was peeking through the window and was being reflected off on many surfaces. In front of the camera were diffraction glasses, the tool responsible for the long, bright streaks seen. Between the sunglight, interior bus safety lights and reflective surfaces, colored diffraction patterns are formed from the glasses' film manipulating the initial light, changing the way the observer percieves the outside world.



Rainbow

Glory

- Most likely known as the most famous and beloved optical phenomenon, rainbows are a diffraction of light caused by a transparent, shaped material. The material separates the visible white light into a spectrum of individual wavelengths. In nature, rainbows brighten up cloudy skies when the light from the sun is unwavering and direct and the atmosphere is riddled with water molecules. A man-made approach can be achieved with a light beam and a prism (Newton's famous prism experiment).
- The left picture isn't exactly a rainbow but is a different effect that is unique to airplanes. This is called a glory; a production of a halo-like effect when sunlight interacts with water molecules in the atmosphere. The circular rainbow is from the light scattering backwards, unlike regular refraction and reflection. This can only be accomplished with special weather conditions: partly cloudy (for the glory to impose on), a humid atmosphere, and the beaming sun.



Psychology of Light Elements

Depth Perception

Pictorial Depth Cues - Occlusion



- Depth perception is the relationship between our three-dimensional world, two-dimensional retinal image, and our own visual experience (perception). These elements depend on varying properties, such as the size, shape, and color of the environment. Pictorial depth cues are a type of monocular cue, or what we can see with one eye. Specifically, pictorial depth cues are what can be observed about a picture.
- Both images illustrate pictorials depth cues to a subjective, yet recognizable degree. The star of the show, or the cue most distinguishable (especially in the image to the left), is known as occlusion. Occlusion is a type of depth cue that arises when one object covers another with undetermined distances between them. This impacts how the human brain estimates the size of an object vs. the distance of the object from the observer. This uncertainty tricks the brain into believing the picture is a certain depth.

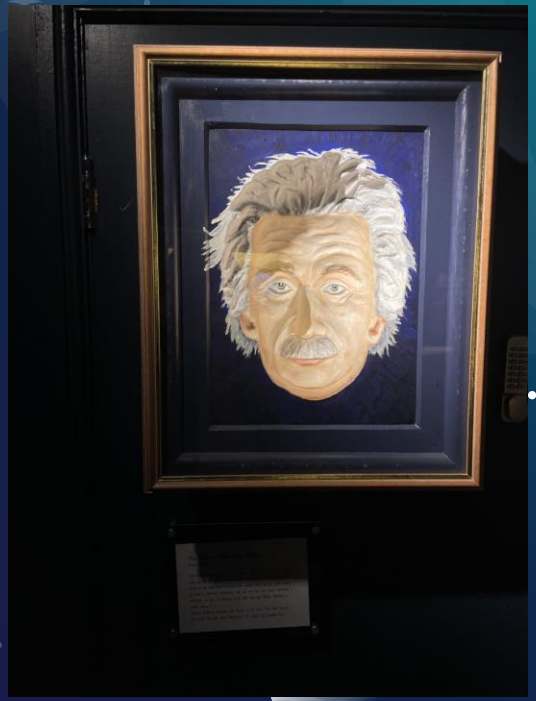


Color & Shadow Perception

Optical Illusion - Shadow Edge



- Color and shadow perception are vital to images that are optical illusions. When the lines or edges on an image go against what appears to be real, the human brain creates a visual explanation for the confusing perspective before it. It is like a mental disguise; or, an artistic, picturesque deception. The ambiguity between the edge definition, illuminated areas, and reflected space controls the level of intensity that an optical illusion can subject its observer to.
- While the image on the left demonstrates color perception, the image that will be focused on is the Einstein Hollow Face Illusion. Despite Einstein's face giving the impression that its surface is flat, or maybe even protruding outward beyond the frame, his face is actually concave. In reality, the shaded areas of the picture aren't darker paints, but are shadows from the light source.





Object Perception

Grouping - Similarity

- Objection perception is the concept of how the human eye can determine certain charactistics of a picture, or even the outside world, through preexisting objects. With object perception, the order of object can double as a depth cue, along with establishing the distance between points in the image. This phenomenon can be explained by Gestalt Laws of Perceptional organization; or to elaborate more indepth, specifically regarding its laws of grouping.
- Under the laws of grouping are a laundry list of perception factors, such as proximity, similarity, good continuation, closure, and common fate. While both images include many combinations of these elements, the image to the left and its relationship with similarity will be focused on. The lampshades are the objects in this case; it can be concluded that these lampshades are identical in level of luminance (little to none) and share a similar color scheme. Along with these similarities, the proximity between the objects is at a minimum.



Faces

Orientation-specific



- Face perception is a skill where the details of a face become identifiable through memorization and objection recognition. It can manifest through three primary features: holistic, orientation specific, and modular. When percieving a face, its characteristics are initially recognized as a “big picture”, rather than the smaller, lesser defining details. This is the idea that smaller parts make a whole, or the whole that the human brain picks up on.
- When growing familiar with a face, that face must be orientation-specific– or to put it simple, it must be upright. For instance, take into consideration the face of the wax figure. The facial features are all upright, not only because that is the nature average person, but due to the way the general outside world is presented as well. As the incoming light enters the eyeball, the retina inverts the image and the gathered visual information is sent back into the brain. Therefore, it is more comfortable and easier for a person to learn a face if it is held upright.

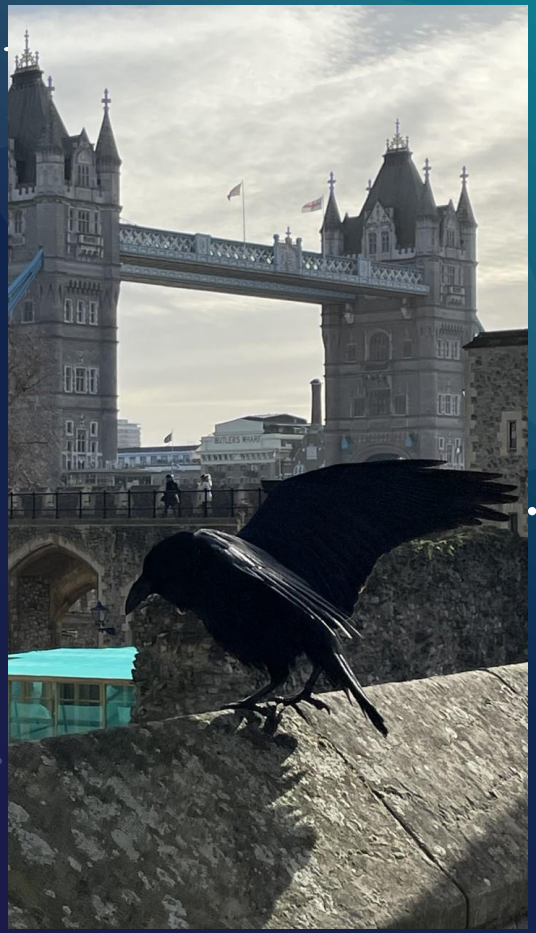


Perception & Action

Effective Action



- Perception and action helps humans grasp a better understanding of our three-dimensional world. It allows the human eye to connect with its surroundings and get an accurate representation of the environment, including objects, space, and other contributing factors. Effective action falls underneath the perception and action umbrella; it is a set of behaviors, such as walking, reaching, jumping, and other actions. As people, we learn to behave with the least amount of processing.
- The picture to the right features a raven, named Jubilee, flapping its wings at the Tower of London. Since the wings are spread apart at an equal distance from each other, it can be assumed that the bird is flapping its wings without the need to actually watch the scenario unfold. There is minimum information needed to arrive to the conclusion of what the bird is doing, because our brains are wired to fill in those blanks.

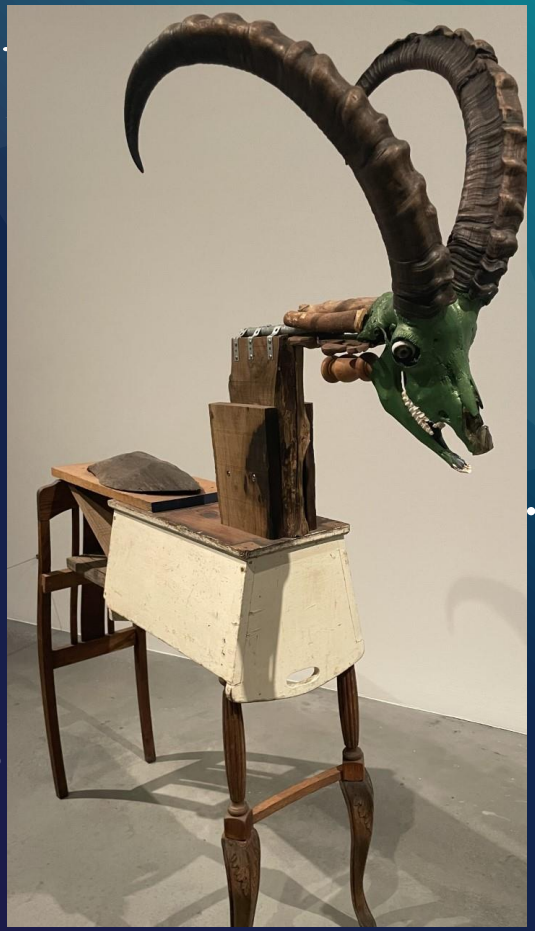




Animal Eyes

Pupils

- Various types of animal eyes are a result of generations upon generations of evolution. Animals have developed their eyes based off their needs for survival, like whether they are the hunted or hunter, how to locate shelter, and how to find a mate. The clarity and brightness of a species' eyesight entirely relies on its size, number of eyes, complexity, placement on head, pigment, and the function of its pupils.
- The tortoises in the picture have wide, pitch black pupils with no sclera exposed. From these observations, it can be concluded that these tortoises are dependent on daylight to illuminate their surroundings (not nocturnal) and perceive objects based on their silhouette, color, and movement. This means that their eyesight may not have the capacity to analyze their environment with exact or strong clarity. Their shells provide protection from predators to accommodate for a lack of speed and strong, accurate eyesight.



Personal Reflection



The Light & Sight courses have taught me to understand my world in a series of explanations rather than purely on circumstance. It reminded me when I first began to take physics courses and how that shaped my overall awareness of what really goes on in the world. Sometimes, what we see is what we make of it and other times, we may not see what is presented to us. It is a question of what is real and what isn't; what we physically see might be real, but uncertainty still lies in our reasoning. That's why science is so fascinating! It is so interesting to see how paradigms and big, wide-spread beliefs over time have shifted and changed in response to ground-breaking discoveries. The connection between what we perceive and why we think we perceive it is so important...it is how we have developed our civilization and culture over many millennium.

SOURCES

[1] Walmsley, Ian. (2015). *Light: A Very Short Introduction*. Oxford University Press.